

EE 473 - HOMEWORK 4

Boran Aybak kılıç
2022401207

$$1) \quad x_1(n) = \{1, 2, 3, 1\}$$

$$x_2(n) = \{4, 3, 2, 2\}$$

$$x_3(m) = \sum_{n=0}^{N-1} x_1(n) x_2((m-n))_N, \quad m=0, \dots, M-1$$

$$x_3(0) = x_1(0)x_2(0) + x_1(1)\underbrace{x_2(-1)}_{x_2(3)} + x_1(2)\underbrace{x_2(-2)}_{x_2(2)} + x_1(3)\underbrace{x_2(-3)}_{x_2(1)}$$

$$= 1 \cdot 4 + 2 \cdot 2 + 3 \cdot 2 + 1 \cdot 3 = 17$$

$$x_3(1) = 1 \cdot 3 + 2 \cdot 4 + 3 \cdot 2 + 1 \cdot 2 = 19 \Rightarrow \text{shift } x_2(-n) \text{ to the right by 1.}$$

$$x_3(2) = 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + 1 \cdot 2 = 22 \Rightarrow \text{" " by 2}$$

$$x_3(3) = 1 \cdot 2 + 2 \cdot 2 + 3 \cdot 3 + 1 \cdot 4 = 19 \Rightarrow \text{" " by 3}$$

$$x_3(n) = \{17, 19, 22, 19\}$$

$$2) \quad X_1(k) = \sum_{n=0}^{N-1} x_1(n) e^{-j2\pi nk/N}$$

$$X_1(k) = \sum_{n=0}^3 x_1(n) (-j)^{nk} = 1 + 2(-j)^k + 3(-1)^k + 1(+j)^k$$

$$\left. \begin{aligned} X_1(0) &= 7, \\ X_1(1) &= 1 - 2j - 3 + j = -2 - j \\ X_1(2) &= 1 - 2 + 3 - 1 = 1, \\ X_1(3) &= 1 + 2j - 3 - j = -2 + j \end{aligned} \right\} X_1(k) = \{7, -2-j, 1, -2+j\}$$

$$X_2(k) = \sum_{n=0}^3 x_2(n) (-j)^{nk} = 4 + 3(-j)^k + 2(-1)^k + 2(+j)^k$$

$$\left. \begin{aligned} X_2(0) &= 11, \\ X_2(1) &= 4 - 2 - 3j + 2j = 2 - j \\ X_2(2) &= 4 - 3 + 2 - 2 = 1, \\ X_2(3) &= 4 + 3j - 2 - 2j = 2 + j \end{aligned} \right\} X_2(k) = \{11, 2-j, 1, 2+j\}$$

2)

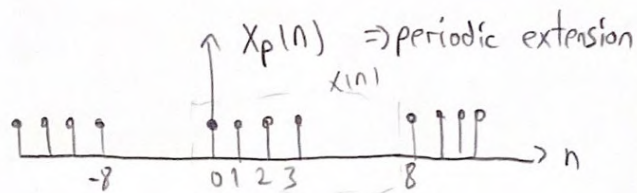
$$X_3(k) = X_1(k) X_2(k) = \{77, -5, 1, -5\}$$

$$X_3(n) = \frac{1}{4} \sum_{k=0}^3 X_3(k) \underbrace{(j)^{kn}}_{e^{j2\pi nk/4}} = \frac{1}{4} [77 - 5(j)^n + (-1)^n - 5(-j)^n]$$

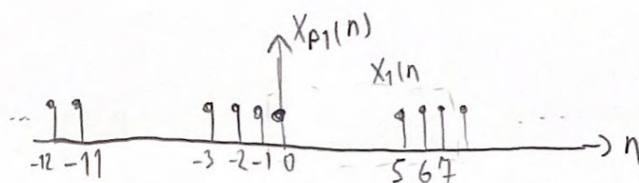
$$\left. \begin{aligned} X_3(0) &= \frac{68}{4} = 17, & X_3(1) &= \frac{76}{4} = 19 \\ X_3(2) &= \frac{88}{4} = 22, & X_3(3) &= \frac{76}{4} = 19 \end{aligned} \right\} \boxed{X_3(n) = \{17, 19, 22, 19\}}$$

3)

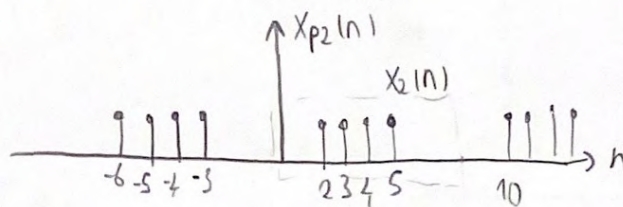
$$X(n) = \begin{cases} 1, & 0 \leq n \leq 3 \\ 0, & 4 \leq n \leq 7 \end{cases}$$



$$X_1(n) = \begin{cases} 1, & n=0 \\ 0, & 1 \leq n \leq 4 \\ 1, & 5 \leq n \leq 7 \end{cases}$$



$$X_2(n) = \begin{cases} 0, & 0 \leq n \leq 1 \\ 1, & 2 \leq n \leq 5 \\ 0, & 6 \leq n \leq 7 \end{cases}$$



* Notice $X_{p1}(n) = X_p(n+3) = X_p(n-5)$ where $X_{p1}(n) = \sum_{l=-\infty}^{\infty} X(n-8l)$

and $X_{p2}(n) = X_p(n-2) = X_p(n+6)$ where $X_{p2}(n) = \sum_{l=-\infty}^{\infty} X_2(n-8l)$

Then by time-shift property:

$$\begin{aligned} X_1(k) &= X(k) e^{-j\frac{2\pi 5k}{8}} = X(k) e^{-j\frac{5\pi k}{4}} \\ X_2(k) &= X(k) e^{-j\frac{2\pi 2k}{8}} = X(k) e^{-j\frac{\pi k}{2}} \end{aligned}$$

→ Since DFT's of periodic extension (X_p) is equal to the DFT of the signal itself.

$$4) \quad a) \quad x(n) = \delta(n) \Rightarrow X(k) = \sum_{n=0}^{N-1} \delta(n) e^{-j2\pi nk/N} = \boxed{1, \quad 0 \leq k \leq N-1}$$

$$b) \quad x(n) = \delta(n-n_0) \Rightarrow X(k) = \sum \delta(n-n_0) e^{-j2\pi nk/N} = \boxed{e^{-j2\pi n_0 k/N}, \quad 0 \leq k \leq N-1}$$

$$c) \quad x(n) = a^n$$

$$X(k) = \sum_{n=0}^{N-1} a^n e^{-j\frac{2\pi k}{N}n} = \sum_{n=0}^{N-1} (a e^{-j\frac{2\pi k}{N}})^n = \frac{1 - a^N e^{-j\frac{2\pi k}{N}N}}{1 - a e^{-j\frac{2\pi k}{N}}}$$

$$= \boxed{\frac{1 - a^N}{1 - a e^{-j2\pi k/N}}}$$

$$d) \quad x(n) = \begin{cases} 1, & 0 \leq n \leq N/2-1 \\ 0, & N/2 \leq n \leq N-1 \end{cases}$$

$$X(k) = \sum_{n=0}^{N/2-1} 1 \cdot e^{-j\frac{2\pi k}{N}n} = \frac{1 - e^{-j\frac{2\pi k}{N} \cdot \frac{N}{2}}}{1 - e^{-j\frac{2\pi k}{N}}} = \frac{1 - e^{-j\pi k}}{1 - e^{-j\frac{2\pi k}{N}}} = \boxed{\frac{1 - (-1)^k}{1 - e^{-j\frac{2\pi k}{N}}}}$$

$$e) \quad x(n) = e^{j\frac{2\pi}{N}k_0 n}$$

$$X(k) = \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-k_0)n} = \begin{cases} N, & k=k_0 \\ 0, & k \neq k_0 \end{cases} = \boxed{N \delta(k-k_0)}$$

$\Rightarrow$ For any integer other than 0, sum $\sum_{k=0}^{N-1} e^{-j\frac{2\pi}{N}k} = 0$

$$d) \quad x(n) = \cos\left(\frac{2\pi}{N}k_0 n\right) = \frac{1}{2} e^{j\frac{2\pi}{N}k_0 n} + \frac{1}{2} e^{-j\frac{2\pi}{N}k_0 n}$$

$\Rightarrow$ From above result, we know that. $\text{DFT} \{e^{-j\frac{2\pi}{N}k_0 n}\} = N \delta(k-k_0)$

Thus $\boxed{X(k) = \frac{N}{2} \delta(k-k_0) + \frac{N}{2} \delta(k+k_0)}$ with $\delta(k+k_0) = \delta(k-N+k_0)$

$$g) X(n) = \sin \frac{2\pi n}{N} k_0 n = \frac{1}{2j} e^{j \frac{2\pi n}{N} k_0 n} - \frac{1}{2j} e^{-j \frac{2\pi n}{N} k_0 n}, \quad 0 \leq n \leq N-1$$

* Exactly as last option:

$$X(k) = \frac{N}{2j} [\delta(k-k_0) - \delta(k-N+k_0)]$$

$$h) X(n) = \begin{cases} 1, & n \text{ even} \\ 0, & n \text{ odd} \end{cases}, \quad 0 \leq n \leq N-1$$

$$X(k) = \sum_{\substack{n=0 \\ n \text{ even}}}^{N-1} 1 \cdot e^{-j \frac{2\pi}{N} k n} \xrightarrow[\substack{* \text{ ASSUME} \\ N \text{ is odd (for now)}}]{\text{let } n=2m} X(k) = \sum_{m=0}^{\frac{N-1}{2}} (e^{-j \frac{2\pi}{N} 2k})^m$$

$$X(k) = \frac{1 - e^{-j \frac{2\pi}{N} \cdot 2k \left(\frac{N+1}{2}\right)}}{1 - e^{-j \frac{4\pi}{N} k}} = \frac{1 - e^{-j 2\pi k} e^{-j \frac{2\pi}{N} k}}{1 - e^{-j \frac{4\pi}{N} k}} = \frac{1 - e^{-j \frac{2\pi}{N} k}}{(1 - e^{-j \frac{2\pi}{N} k})(1 + e^{-j \frac{2\pi}{N} k})}$$

$$X(k) = \boxed{\frac{1}{1 + e^{-j \frac{2\pi}{N} k}}} \quad \text{for odd } N$$

For even N we would not have the last component in the summation

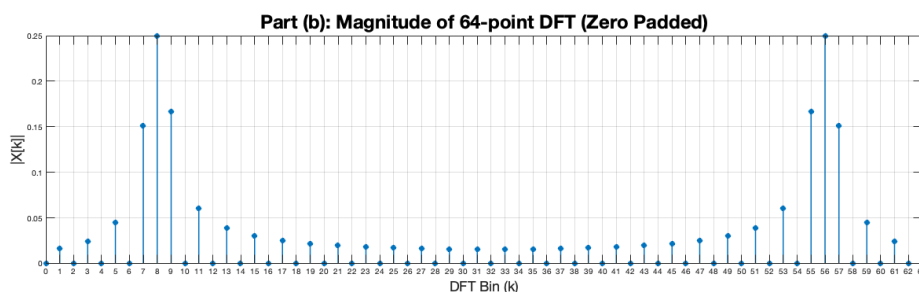
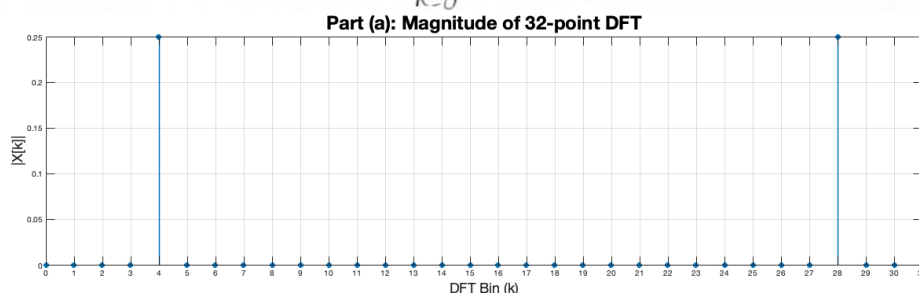
$$\hookrightarrow X(k) = \frac{1}{1 + e^{-j \frac{2\pi}{N} k}} - e^{-j \frac{2\pi}{N} (N-1)k} = \frac{1}{1 + e^{-j \frac{2\pi}{N} k}} - e^{+j \frac{2\pi}{N} k} = \frac{-e^{+j \frac{2\pi}{N} k}}{1 + e^{-j \frac{2\pi}{N} k}}$$

5)

a) $x(n) = \frac{1}{64} \cos(\pi n/4)$ for $0 \leq n \leq 31$, $X(k) = \frac{1}{64} \sum_{n=0}^{31} x(n) e^{-j \frac{2\pi}{32} nk} \Rightarrow 32 \text{ point DFT}$
 $= \frac{1}{128} \sum_{n=0}^{31} e^{-j \frac{2\pi}{32} (k-4)n} + e^{-j \frac{2\pi}{32} (k+4)n} = \frac{1}{4} [\delta(k-4) + \delta(k+4)]$

b) For the 64 point DFT, define $x'(n) = \begin{cases} x(n), & 0 \leq n \leq 31 \\ 0, & 32 \leq n \leq 63 \end{cases}$

$$X'(k) = \sum_{n=0}^{63} x(n) e^{-j \frac{2\pi}{64} nk} = \frac{1}{64} \sum_{n=0}^{31} \cos(\pi n/4) e^{-j \frac{2\pi}{64} nk}$$



c) In part a), we have $N=32$. Since signal period is 8, our window contains exactly 4 full cycles. Because window length is an integer multiple of signal period, there is no spectral leakage and signal is perfectly represented with 32-point DFT

In part b), however we need a 64-point signal to take 64-point DFT

Thus we pad $x(n)$ with zeros $\Rightarrow x'(n) = \begin{cases} x(n), & 0 \leq n \leq 31 \\ 0, & 32 \leq n \leq 63 \end{cases}$

$x(n) = x'(n) W_{32}(n)$ where $W_{32}(n) = \begin{cases} 1, & 0 \leq n \leq 31 \\ 0, & 32 \leq n \leq 63 \end{cases} \Rightarrow \text{rectangular window of length 32.}$

The we will have $\text{DFT}_{64}\{x(n)\} = \text{DFT}_{64}\{x'(n)\} * \text{DFT}_{64}\{W_{32}(n)\}$

$$\text{DFT}_{64}\{x'(n)\} = \frac{1}{128} \sum_{n=0}^{31} e^{-j\frac{2\pi}{64}n(k-8)} + e^{-j\frac{2\pi}{64}n(k+8)}$$

$$= \frac{1}{4} [\delta(k-8) + \delta(k+8)]$$

$$\begin{aligned} \text{DFT}_{64}\{W_{64}(n)\} &= \sum_{n=0}^{31} e^{-j\frac{2\pi}{64}nk} = \frac{1 - e^{-j\frac{2\pi}{64}k32}}{1 - e^{-j\frac{2\pi}{64}k}} = \frac{e^{-j\frac{2\pi}{64}16k} \sin(\frac{\pi k}{2})}{e^{-j\frac{\pi}{64}k} \sin(\frac{\pi k}{64})} \\ &= e^{-j\frac{31\pi k}{64}} \cdot \frac{\sin(\pi k/2)}{\sin(\pi k/64)} \end{aligned}$$

$\Rightarrow$ Thus 64-point DFT will be as if 2 Dirac Deltas as $k=8$ and $k=56$ is convolved with $e^{-j\frac{31\pi k}{64}} \frac{\sin(\pi k/2)}{\sin(\pi k/64)}$

6) $x(n) = \{ \underset{\uparrow}{1}, 2, 3, 4, 4, 3, 2, 1 \}$, $h(n) = \{ \underset{\uparrow}{1}, 1, 1 \}$

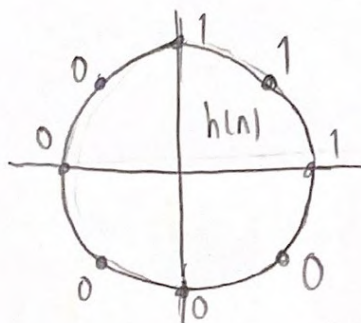
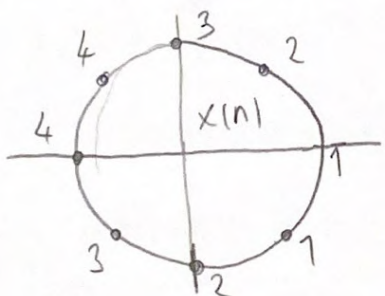
6.1) $y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k) = \sum_{k=0}^{\infty} x(k) h(n-k)$

$$\left. \begin{array}{ll} y(0) = 1 & y(5) = 4+4+3 \\ y(1) = 1+2 & y(6) = 4+3+2 \\ y(2) = 1+2+3 & y(7) = 3+2+1 \\ y(3) = 2+3+4 & y(8) = 2+1 \\ y(4) = 3+4+4 & y(9) = 1 \end{array} \right\}$$

$$y(n) = \{1, 3, 6, 9, 11, 11, 9, 6, 3, 1\}$$

linear conv.

6.2) Pad $h(n)$ with zeros to have length 8 $\Rightarrow h(n) = \{1, 1, 1, 0, 0, 0, 0, 0\}$



$\rightarrow y(n) = x(n) \otimes h(n) = x'(n) * h'(n)$ where $x'(n)$ & $h'(n)$ are 8-point periodic extensions.

$$\left. \begin{array}{ll} y(0) = 2+1+1 & y(4) = 3+4+4 \\ y(1) = 1+1+2 & y(5) = 4+4+3 \\ y(2) = 1+2+3 & y(6) = 4+3+2 \\ y(3) = 2+3+4 & y(7) = 3+2+1 \end{array} \right\}$$

$$y(n) = \{4, 4, 6, 9, 11, 11, 9, 6\}$$

circular conv.

6.3)

$$a) X_8(k) = \sum_{n=0}^7 x(n) e^{-j \frac{2\pi}{8} nk} = 1 + 2e^{-j \frac{\pi}{4} k} + 3e^{-j \frac{\pi}{2} k} + 4e^{-j \frac{3\pi}{4} k} + 4e^{-j \pi k} + 3e^{-j \frac{5\pi}{4} k} + 2e^{-j \frac{3\pi}{2} k} + 1e^{-j \frac{7\pi}{4} k}$$

$$H_8(k) = \sum_{n=0}^7 h(n) e^{-j \frac{2\pi}{8} nk} = 1 + 1e^{-j \frac{\pi}{4} k} + 1e^{-j \frac{\pi}{2} k}$$

Using MATLAB $\Rightarrow Y_8(k) = X_8(k) H_8(k) = \{60, -14.07 + 5.83j, 0, 0, 0, 0.071 - 0.141j, 0, -14.07 - 5.8284j\}$

$$y_8(n) = \{4, 4, 6, 9, 11, 11, 9, 6\}$$

$$b) X_{10}(k) = \sum_{n=0}^9 x(n) e^{-j \frac{2\pi}{10} nk}, \quad H_8(k) = \sum_{n=0}^9 h(n) e^{-j \frac{2\pi}{10} nk}$$

$\hookrightarrow$ essentially up to $n=7$ $\hookrightarrow$ essentially up to $n=2$

Using MATLAB $\Rightarrow Y_{10}(k) = H_{10}(k) X_{10}(k) = \{60, -24.8 - 8.05j, 0, -0.2 - 0.277j, 0, 0, 0, -0.2 + 0.277j, 0, -24.8 + 8.05j\}$

$$y_{10}(n) = \{1, 3, 6, 9, 11, 11, 9, 6, 3, 1\}$$

$$c) \text{ Similarly } X_{16}(k) = \sum_{n=0}^{15} x(n) e^{-j \frac{\pi}{8} nk}, \quad H_{16}(k) = \sum_{n=0}^{15} h(n) e^{-j \frac{\pi}{8} nk}$$

Using MATLAB $\Rightarrow y_{16}(n) = \text{IDFT}_{16}\{X_{16}(k) H_{16}(k)\} = \{1, 3, 6, 9, 11, 9, 6, 3, 1, 0, 0, 0, 0, 0, 0, 0\}$

d) $\rightarrow$ 8-point DFT + multiplication + IDFT corresponds to circular convolution in part 1)

$\rightarrow$ When DFT length is chosen to be less than required convolution length of 10, time-domain signal is an aliased version of linear convolution as in part a)

$\rightarrow$ When 10-point DFT is used, we exactly get the linear convolution in part 2)

$\rightarrow$ When N is larger than there is no aliasing but resulting signal is padded with $N-10$ zeros as in part c)

Code for calculating DFT and IDFTs

```
1  % Define the sequences
2  x = [1, 2, 3, 4, 4, 3, 2, 1];
3  h = [1, 1, 1];
4
5  % --- Calculate X_N(k) ---
6  X_8 = fft(x, 8);
7  X_10 = fft(x, 10);
8  X_16 = fft(x, 16);
9
10 % --- Calculate H_N(k) ---
11 H_8 = fft(h, 8);
12 H_10 = fft(h, 10);
13 H_16 = fft(h, 16);
14
15 disp('X_8(k):'); disp(X_8.*H_8);
16 disp('H_8(k):'); disp(ifftt(X_8.*H_8));
17
18 disp('X_10(k):'); disp(X_10.*H_10);
19 disp('H_10(k):'); disp(ifftt(X_10.*H_10));
20
21 disp('X_16(k):'); disp(X_16.*H_16);
22 disp('H_16(k):'); disp(ifftt(X_16.*H_16));
```